Worksheet 4-5 C Related Rates #3
AP Calculus Name_____

Directions: Read the following statements. Select the choice that correctly **interprets** the given information.

1. Water is draining out of a conical tank with height h and radius r at a rate of 8 cubic liters per second.

A) $\frac{dh}{dt} = 8$ B) $\frac{dh}{dt} = -8$ C) $\frac{dV}{dt} = -8$ D) $\frac{dV}{dt} = 8$

2. The area of a circle with radius r is increasing. Find the rate that the radius is changing when the circumference of the circle is increasing at a rate of 4 inches per second.

A) $\frac{dA}{dt} = 4$ B) $\frac{dC}{dt} = 4$ C) $\frac{dr}{dt} = 4$ D) $\frac{dC}{dt} = 4\pi$

3. A triangle with base b and height h is expanding such that its area is increasing at the rate $4\ \mathrm{m}^2/\mathrm{s}$.

A) $\frac{dA}{dt} = 4$ B) $\frac{db}{dt} = 4$ C) $\frac{dh}{dt} = 4$ D) $\frac{dt}{dA} = 4$

4. A spherical rock erodes over time due to winds and rain. The radius of the rock is changing at a rate of 3 in/year. Find the rate that the volume of the rock is changing when the surface area (A) of the rock is $400\ \mathrm{in}^2$.

A) $\frac{dA}{dt} = 3$ B) $\frac{dr}{dt} = 3$ C) $\frac{dV}{dt} = -3$ D) $\frac{dr}{dt} = -3$

Directions: Solve the following problems. Show all work clearly and neatly ON NOTEBOOK PAPER.

5. Let s and r be variables such that $2s = \sqrt{3r+1}$. Find $\frac{dr}{dt}$ when $\frac{ds}{dt} = 5$ and $s = 2$.

6. A particle is moving along the graph of $xy^2 = 36$ where $\frac{dy}{dt} = -2$. Find $\frac{dx}{dt}$ when $y = -3$.

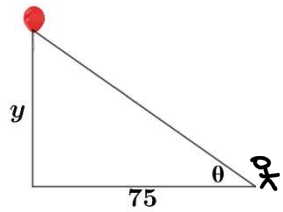
7. Water is running out of a conical funnel at the rate of 1 cubic inch per second. If the radius of the base of the funnel is 4 in. and the altitude is 8 in., find the rate at which the water level is dropping when it is 2 in. from the top. (Volume: $V = \frac{1}{3}\pi r^2 h$)

8. A train, starting at 11 a.m., travels east at 45 mi/hr. while another, starting at noon from the same point, travels south at 60 mi/hr. How fast are they separating at 3 p.m.?

9. A boy is flying a kite at a height of 150 ft. If the kite moves horizontally away from the boy at the rate of 20 ft/sec, how fast is the string being paid out when the kite is 250 ft. from him?

10. A man 5 ft. tall walks at the rate 4 ft/sec. directly away from a streetlight which is 20 ft. above the street. At what rate is the length of his shadow changing and at what rate is the tip of his shadow changing?

11. A balloon is rising vertically from the ground at a rate of 3 feet per second. A person is standing 75 feet away from the point on the ground directly below the balloon.

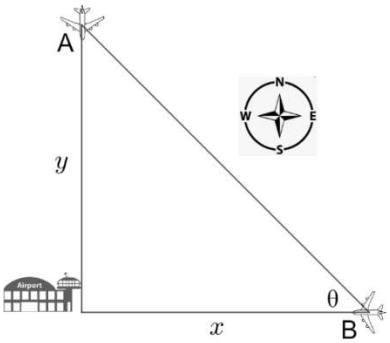


a) Find the rate, in feet per second, that the distance between the person and the balloon is changing when the balloon is 40 feet above the ground.

b) At what rate is angle θ changing, in radians per second, when the balloon is 40 feet above the ground?

c) The balloon continues to rise above the ground until it eventually pops and begins to fall vertically back toward the ground. When the balloon is 100 feet above the ground, the distance between the person and the balloon is decreasing at a rate of 25 feet per second. Find the rate that the height of the balloon is falling when it is 100 feet above the ground.

12. Two planes, A and B, are flying toward an airport. Plane A is traveling due south at a rate of 200 miles per hour. Plane B is traveling due west at a rate of 250 miles per hour. Let x and y represent the distances between the planes and the airport at time t.



a) Find the rate that the distance between plane A and plane B is changing when $x = 400$ mi and $y = 300$ mi. Include units of measure.

b) Let θ be the angle shown in the figure. Find the rate of change of θ when $x = 400$ mi and $y = 300$ mi. Include units of measure.

Answers:

1.) C 2.) D 3.) A 4.) D 5.) $\frac{80}{3}$ 6.) $-\frac{16}{3}$

7.) The water level is dropping at a rate of $\frac{1}{9\pi}$ in/s

8.) The trains are separating at a rate of 74.246 mi/h at 3pm

9.) The string is being paid out at a rate of 16 ft/s

10.) The length of his shadow is increasing at a rate of $\frac{4}{3}$ ft/s; the tip of shadow is changing at a rate of $\frac{16}{3}$ ft/s

11.) a) The distance between the person and the balloon is increasing at a rate of $\frac{24}{17}$ ft/s

b) θ is increasing at a rate of $\frac{9}{289}$ radians/s

c) The balloon is falling at a rate of $\frac{125}{4}$ ft/s

12.) a) The distance between Plane A and Plane B is decreasing at a rate of 320 mi/h

b) θ is decreasing at a rate of $\frac{1}{50}$ radians/hr